

Bayesian Learning (Bayes Theorem)

- Bayes rule / theorem helps to determine the probability of an event with random knowledge
- It is used to calculate the probability of occurring one event while other one already occurred.
- It is a best method to relate the condition probability and marginal probability.
- In simple words, we can say that Bayes theorem helps to contribute more accurate results.
- ~~It is hypo~~
- Bayes theorem is one of the most popular machine learning concepts that helps to calculate the probability of occurring one event with uncertain knowledge while other one has already occurred.

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$P(A/B)$: conditional probability - Posterior Probability

$P(B/A)$: " " - likelihood

$P(A)$: Prior

$P(B)$: Marginal

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \text{ --- (1), } P(B/A) = \frac{P(B \cap A)}{P(A)} \text{ --- (2)}$$

From equation (1)

$$P(A \cap B) = P(A/B) \cdot P(B) \text{ --- (3)}$$

From equation (2)

$$P(B \cap A) = P(B/A) \cdot P(A) \text{ --- (4)}$$

From equation (3) & (4), $P(A \cap B)$ and $P(B \cap A)$ are equal so,

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

$$\therefore P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

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Naive Bayes classifier:

- ↳ Supervised ML alg.
- ↳ Based on Bayes theorem
- ↳ Probabilistic classifiers
- ↳ Assum. features are conditionally independent w.r to given output / target class.

Bayes theorem example.

2 Red	$E_1 = \frac{1}{2}, \frac{2}{5} (R)$
3 Black	

4 Red	$E_2 = \frac{1}{2}, \frac{4}{7} (R)$
3 Black	

$$\frac{1}{2} \times \frac{2}{5} + \frac{1}{2} \times \frac{4}{7}$$

Bag 1 contain 2 Red and 3 black balls. Bag 2 contain 3 Red and 4 black balls. One ball is drawn at random from one of these bags and it's red. Find the probability that it is drawn from bag 1.

A given that ball is Red, whether find the probability for bag 1. Thus we have find (B_1/R) which can be solved by Bayes theorem.

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

→ Prior

posterior probability

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Bayes theorem ~~derive~~ based on conditional probability

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \quad \text{--- (1)}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} \quad \text{--- (2)}$$

where $P(A \cap B)$: probability for common to both A and B

$P(A/B)$: probability of occurrence of A given B has occurred

$P(B)$: probability of B

from (1) & (2)

$$P(A/B) \cdot P(B) = P(A \cap B) \quad \text{--- (3)}$$

$$P(B/A) \cdot P(A) = P(A \cap B) \quad \text{--- (4)}$$

from (3) & (4)

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

$$\therefore P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

This is Bayes theorem

Date / /



Naive Bayes

$$P(Y/X) = \frac{P(X/Y) \cdot P(Y)}{P(X)}, \quad P(N/X) = \frac{P(X/N) \cdot P(N)}{P(X)}$$

→ Yes

$$P(Y/x_1, x_2, x_3, \dots, x_n) = \frac{P(x_1/Y) \cdot P(x_2/Y) \cdot \dots \cdot P(x_n/Y) \cdot P(Y)}{P(x_1) \times P(x_2) \cdot \dots \cdot P(x_n)}$$

→ No

$$P(N/x_1, x_2, x_3, \dots, x_n) = \frac{P(x_1/N) \cdot P(x_2/N) \cdot \dots \cdot P(x_n/N) \cdot P(N)}{P(x_1) \cdot P(x_2) \cdot \dots \cdot P(x_n)}$$

Naïve Bayes

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Naïve Bayesian Algorithm

The dataset consists of n dimension attribute along with associated class label. Each sample is represented as n dimensional attribute vector $X = x_1, x_2, \dots, x_n$ and has associated label as $C_1, C_2, \dots, C_n$

Step 1: A sample is set to belong to class $C_1, C_2, \dots, C_n$ for which it has maximum value of posterior probability conditioned on X

$$P(C_i|X) > P(C_j|X)$$

It is computed as

$$P(C_i|X) = \frac{P(X|C_i)P(C_i)}{P(X)}$$

Step 2: Since $P(X)$ is constant for all classes C_i , so maximization of $P(X|C_i)P(C_i)$ needs to be done. If $P(C_i)$ is not known, it is considered that each of the class is equally probable and only $P(X|C_i)$ needs to be maximized.

Compute posterior probability conditioned on C_i

$$\begin{aligned} P(X|C_i) &= \prod_{k=1}^n P(x_k|C_i) \\ &= P(x_1|C_i) * P(x_2|C_i) * \dots * P(x_n|C_i) \end{aligned}$$

Here $x_1, x_2, \dots, x_n$ denotes the feature vector

Naïve Bayes

Step 3: Compute the value of A_k for each of the feature is either categorical or continuous. For categorical value.

$$P(x_k|C_i) = \frac{\text{no. of tuples of class } C_i \text{ with } x_k \text{ for } A_k \text{ Dataset}}{\text{Total no. of tuples of } C_i \text{ in Dataset}}$$

If the value of A_k is continuous then Gaussian distribution is used to find the value of $P(x_k, C_i)$

$$P(x_k | C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i}) = \frac{1}{\sqrt{2\pi}\sigma_{C_i}} * e^{\frac{-(x-\mu_{C_i})^2}{2\sigma_{C_i}^2}}$$

Here μ_{C_i} mean of attribute value A_k for class C_i

σ_{C_i} standard deviation of attribute value A_k for class C_i

The Naive Bayes Classifier is a supervised machine learning algorithm used for classification tasks, built on the principles of Bayes' Theorem.

Core Concepts

- Probabilistic:** It calculates the probability of a data point belonging to a specific class and assigns it to the class with the highest probability.
- The "Naive" Assumption:** It assumes that all features (variables) are completely independent of each other. In reality, features often influence one another, but the algorithm ignores this to simplify calculations.
- Use Cases:** Spam filtering, sentiment analysis, document categorization, and fast baseline predictions.

The Math

Naive Bayes relies on Bayes' Theorem to calculate the posterior probability $P(C|X)$ —the probability of class C given feature X :

$$P(C|X) = [P(X|C) * P(C)] / P(X)$$

- $P(C)$: Prior probability (how often the class occurs overall).
- $P(X|C)$: Likelihood (how often the feature appears in that specific class).
- $P(X)$: Evidence (often ignored in classification because it is the same across all classes).

Short Example: Spam Filter

Imagine a dataset of 10 emails categorized as Spam or Not Spam. We want to predict if a new email containing the word "Free" is Spam.

Category	Total Emails	Emails containing "Free"
Spam	6	5
Not Spam	4	1

Step 1: Calculate Prior Probabilities

- $P(\text{Spam}) = 6 / 10 = 0.60$
- $P(\text{Not Spam}) = 4 / 10 = 0.40$

Step 2: Calculate Likelihoods

- $P(\text{"Free"} | \text{Spam}) = 5 / 6 \approx 0.83$
- $P(\text{"Free"} | \text{Not Spam}) = 1 / 4 = 0.25$

Step 3: Calculate Posteriors (Numerator only)

Since $P(\text{"Free"})$ is the same for both calculations, we only need to compare the numerators:

- Spam Score:** $P(\text{"Free"} | \text{Spam}) * P(\text{Spam}) = (5/6) * 0.60 = 0.50$
- Not Spam Score:** $P(\text{"Free"} | \text{Not Spam}) * P(\text{Not Spam}) = 0.25 * 0.40 = 0.10$

Conclusion

Because $0.50 > 0.10$, the algorithm classifies the new email as **Spam**.

Naive Bayes Classifier:

Fruit = { yellow, sweet, long }

Fruit	yellow	sweet	long	total
Orange	350	450	0	650
Banana	400	300	350	1050
others	50	100	50	200
Total	800	850	400	2050

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$$P(\text{yellow/orange}) = \frac{P(\text{orange/yellow}) \cdot P(\text{yellow})}{P(\text{orange})}$$

$$= \frac{\frac{350}{800} \times \frac{800}{2050}}{\frac{650}{2050}}$$

$$= 0.53$$

Similarly

$$P(S/O) = 0.69$$

$$P(L/O) = 0$$

Artificial Neural Network:

↳

$$P(\text{Fruit}/\text{orange}) = 0.5 \times 0.69 \times 0 = 0$$

$$P(\text{Fruit}/\text{Banana}) = 1 \times 0.75 \times 0.87 = 0.65$$

$$P(\text{Fruits}/\text{other}) = 0.33 \times 0.66 \times 0.33 = 0.072$$

Fruit = Banana

Play-tennis example: estimating $P(x_i|C)$

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P
sunny	mild	normal	true	P
overcast	mild	high	true	P
overcast	hot	normal	false	P
rain	mild	high	true	N

$$P(p) = 9/14$$

$$P(n) = 5/14$$

outlook	
$P(\text{sunny} p) = 2/9$	$P(\text{sunny} n) = 3/5$
$P(\text{overcast} p) = 4/9$	$P(\text{overcast} n) = 0$
$P(\text{rain} p) = 3/9$	$P(\text{rain} n) = 2/5$
temperature	
$P(\text{hot} p) = 2/9$	$P(\text{hot} n) = 2/5$
$P(\text{mild} p) = 4/9$	$P(\text{mild} n) = 2/5$
$P(\text{cool} p) = 3/9$	$P(\text{cool} n) = 1/5$
humidity	
$P(\text{high} p) = 3/9$	$P(\text{high} n) = 4/5$
$P(\text{normal} p) = 6/9$	$P(\text{normal} n) = 2/5$
windy	
$P(\text{true} p) = 3/9$	$P(\text{true} n) = 3/5$
$P(\text{false} p) = 6/9$	$P(\text{false} n) = 2/5$

Naive Bayesian Classifier (II)

- z Given a training set, we can compute the probabilities

Outlook	P	N		Humidity	P	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		true	3/9	3/5
mild	4/9	2/5		false	6/9	2/5
cool	3/9	1/5				

Play-tennis example: classifying X

z An unseen sample $X = \langle \text{rain, hot, high, false} \rangle$

z $P(X|p) \cdot P(p) =$

$$P(\text{rain}|p) \cdot P(\text{hot}|p) \cdot P(\text{high}|p) \cdot P(\text{false}|p) \cdot P(p) = \\ 3/9 \cdot 2/9 \cdot 3/9 \cdot 6/9 \cdot 9/14 = 0.010582$$

z $P(X|n) \cdot P(n) =$

$$P(\text{rain}|n) \cdot P(\text{hot}|n) \cdot P(\text{high}|n) \cdot P(\text{false}|n) \cdot P(n) = \\ 2/5 \cdot 2/5 \cdot 4/5 \cdot 2/5 \cdot 5/14 = 0.018286$$

z Sample X is classified in class n (don't play)